

TRAN BAO PHUC LONG

AI Engineer
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LinkedIn — GitHub

SUMMARY

Computer Science student and Algorithm Engineer with hands-on experience building, training, and evaluating ML and deep learning pipelines. Skilled in Python, C++, PyTorch, model inference, and performance evaluation. Interested in AI systems, model optimization, and efficient deep learning execution on specialized hardware.

EDUCATION

HCM City University of Technology - VNU-HCM — Ho Chi Minh City
Major in Computer Science — Sep 2023 – Present
GPA: 3.4/4.0, IELTS 7.0

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL
- **Machine Learning & Deep Learning:** PyTorch, LightGBM, XGBoost, Scikit-learn, PyPOTS, MLflow
- **Data Engineering & Processing:** Polars, Pandas, NumPy, Dagster, ETL Pipelines, Data Imputation

COMPETITIONS & AWARDS

- **Comprehensive Excellent Student Award** - HCMUT — 2024-2025
- 1st **Runner-Up** - SkyTrack Hackathon 2026 — October 2026
- **Top 3 - SGS Campus** - RMIT Hackathon 2026 — October 2026
- Competed in VNPT AI Hackathon 2025, Datathon by VinUni 2026,...

EXPERIENCE

kSynerX Vietnam — Ho Chi Minh City
Algorithm Engineer — March 2026 – September 2026

- Designed and built an end-to-end modular retail forecasting system, from raw data ingestion, high-performance data processing using Polars DataFrames to production API serving across distributed microservices.
- Integrated PyPOTS deep learning architectures within the pipeline to impute missing time-series data and accurately reconstruct latent demand during stockout periods.
- Implemented and benchmarked multi-model forecasting pipelines using LightGBM, XGBoost, and PyTorch, tracked experiments via MLflow, and deployed low-latency inference endpoints using FastAPI.

PROJECTS

Retail Demand Forecasting Engine — Ho Chi Minh City
Personal Project — March 2026

- Developed an end-to-end retail demand forecasting pipeline covering data preparation, feature engineering, stockout-aware demand recovery, model training, and 168-hour forecasting.
- Built and evaluated a hybrid forecasting system using XGBoost and Amazon Chronos-2, comparing models with MAE, RMSE, WAPE, and forecast bias.
- Experimented with SAITS, DLinear, and TimesNet to reconstruct missing and censored demand during stockout periods and improve downstream data quality.
- Deployed XGBoost and Chronos-2 as independent Ray Serve services behind a FastAPI endpoint for model integration and inference.
- **GitHub:** github.com/tbplong/RetailForecast

Vietnamese Question Answering System — Ho Chi Minh City
VNPT AI Hackathon 2025 — December 2025

- Developed an end-to-end Vietnamese question-answering system combining question routing, Retrieval Augmented Generation (RAG), context extraction, and specialized reasoning workflows.
- Designed concurrent retrieval and inference pipelines to process embedding and context requests efficiently while operating within external API limits.

- Integrated knowledge-graph-assisted retrieval and local LLM inference using OpenAI-compatible LM Studio/vLLM APIs for experimentation and system integration.
- Evaluated the solution on benchmark tasks, achieving scores of 0.8125 on the RAG track and 0.7333 on the Multi-domain track.
- **GitHub:** github.com/tbplong/VCM-B

DeepLearningHub - Text Classification Ho Chi Minh City
Course Project April 2026

- Developed an NLP classification pipeline for the 1.4-million-sample Yahoo Answers dataset across 10 topic classes.
- Implemented and trained Bidirectional LSTM and BERT-based models, including tokenization, data loading, training, checkpointing, and inference workflows.
- Evaluated models using accuracy, precision, recall, F1-score, inference latency, parameter count, and model size to compare predictive performance and operational trade-offs.
- **GitHub:** github.com/nat50/DeepLearningHub
- **Web Link:** nat50.github.io/DeepLearningHub/